B1.2 Interoperability

This section describes the interoperability of AXI and AXI4-Lite masters and slaves. Table B1-2 shows the possible combinations of interface, and indicates that the only case requiring special consideration is an AXI master connecting to an AXI4-Lite slave.

Table B1-2 Full AXI and AXI4-Lite interoperability

Master	Slave	Interoperability
AXI	AXI	Fully operational.
AXI	AXI4-Lite	AXI ID reflection is required. Conversion might be required.
AXI4-Lite	AXI	Fully operational.
AXI4-Lite	AXI4-Lite	Fully operational.

B1.2.1 Bridge requirements of AXI4-Lite slaves

As Table B1-2 shows, the only interoperability case that requires special consideration is the connection of an AXI4-Lite slave interface to a full AXI master interface.

This connection requires AXI ID reflection. The AXI4-Lite slave must return the AXI ID associated with the address of a transaction with the read data or write response for that transaction. This is required because the master requires the returning ID to correctly identify the transaction response.

If an implementation cannot ensure that the AXI master interface only generates transactions in the AXI4-Lite subset, then some form of adaptation is required. See *Conversion, protection, and detection* on page B1-127.

B1.2.2 Direct connection requirements of AXI4-Lite slaves

An AXI4-Lite slave can be designed to include ID reflection logic. This means the slave can be used directly on a full AXI connection, without a bridge function, in a system that guarantees that the slave is accessed only by transactions that comply with the AXI4-Lite subset.

Note

This specification recommends that the ID reflection logic uses **AWID**, instead of **WID**, to ensure compatibility with both AXI3 and AXI4.

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